Coordination Games

		Player 2	
		Ballet	Prize Fight
Player 1	Ballet	3, 2	1, 1
	Prize Fight	0, 0	2, 3

- This game has two Nash equilibria (similar as the software launch game)
  - Which NE shall we predict? Extra information is needed …
- An example of "coordination games"
  - Miscoordination occurs if they expect different NE to arise, leading to undesirable outcomes
  - Also, in this game being nice to each other will lead to the worst outcome

Coordination Games

- Many other examples:
  - The guessing game when 2/3 becomes 1
  - New market entry
  - Driving on the left or the right
  - Bank runs
  - Cryptocurrency …
- The existence of multiple NE can sometimes help explain the diversity of social phenomena
- Some laws and institutional designs serve to help people coordinate on a "good" equilibrium

Summary

- Game theory studies decision-making when *strategic interaction* is present
  - *Strategic thinking* is crucial; put yourself into other players' shoes

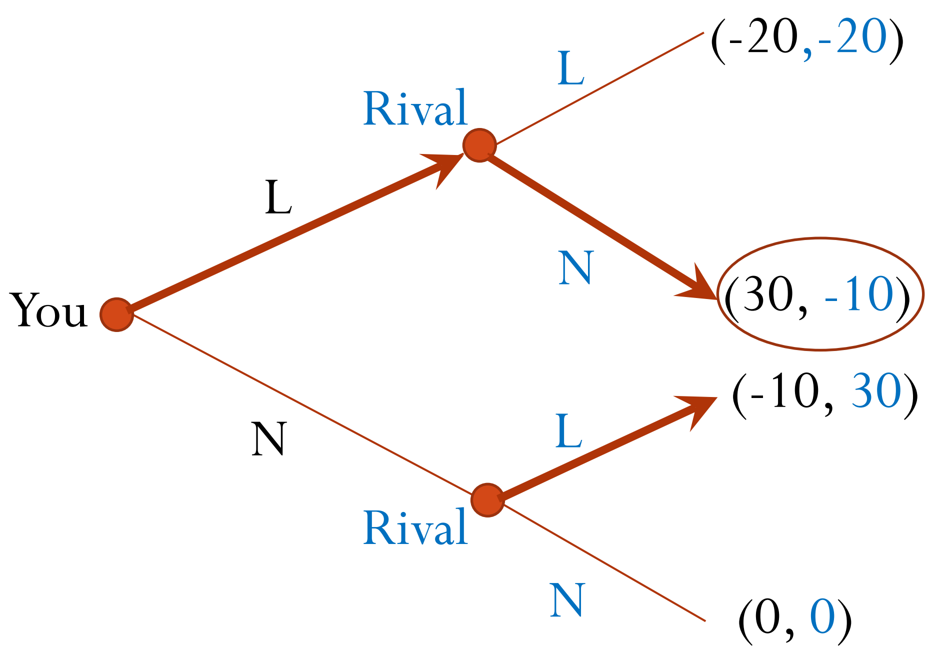
- We use *Nash Equilibrium (NE)* to make predictions
  - Stable situation with no *individual* deviation incentive
  - Method 1: Dominant or dominated strategy (if any)
  - Method 2: Best-response method
  - Method 3: Guess and verify

- Lessons from important games discussed so far:
  - *Prisoner's dilemma*: individual rationality vs collective rationality
  - *Guessing 2/3 of the average*: other players' sophistication; learning
  - *Location choice game*: clustering to attract consumers/voters in the middle
  - *Coordination game*: multiple NE and social phenomenon diversity; institutional design

Sequential-Move Games

Backward Induction

- Use *backward induction* to find the (subgame-perfect) NE in a sequential-move game
  - Look forward and reason backward
  - If you move first, anticipate how your rivals will respond to your choice

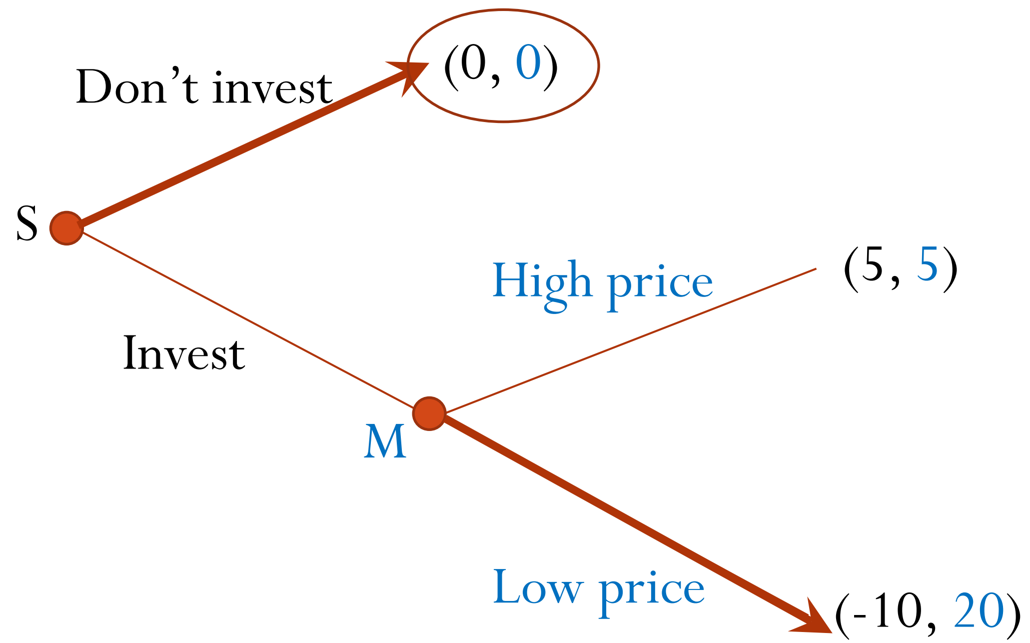


- If you launch, your rival will not
- If you don't launch, your rival will
- Anticipating that, you should launch

The Hold-Up Game

- Upstream supplier S first decides whether to invest in a new plant to produce, say, crankshaft for an automaker M
  - Suppose the crankshaft is designed *specifically* for M's engine, so cannot be used by other automakers
- If the investment is made, M then decides whether to offer a high or low price to S
  - If M offers a high price, M and S each make a net profit $5m
  - If M offers a low price, M earns $20m and S loses $10m
- If no investment, each firm makes a zero profit
- Should S make the investment?

The Hold-Up Game …



Hold-up problem (or opportunism) results in *low efficiency*: the more efficient outcome (5,5) cannot be reached!

The Hold-Up Game …

- Possible remedies?
  - Commitment via contracting
  - Commitment via reputation/long-term relationship
  - Vertical integration

- Other similar games:
  - Borrowing and lending
  - Restaurant service and tips
  - Remote transactions

    (the role of platforms; smart contracts)

  …

Ultimatum Game

- Two players divide $100 prize
- Player A proposes a division plan:

  x for herself/himself, and 100-x for player B
- Player B then decides whether to accept or reject it
- If B rejects it, both players get nothing

  - MW 10:10am (Blue): **m3n4bax34**
  - MW 2:40pm (Silver): **kgp6bax34**
  - TTh 10:10am (Green): **82u7bax34**
  - TTh 1:00pm (Gold): **m8r8bax34**
  - TTh 2:40pm (Red): **3pn9bax34**

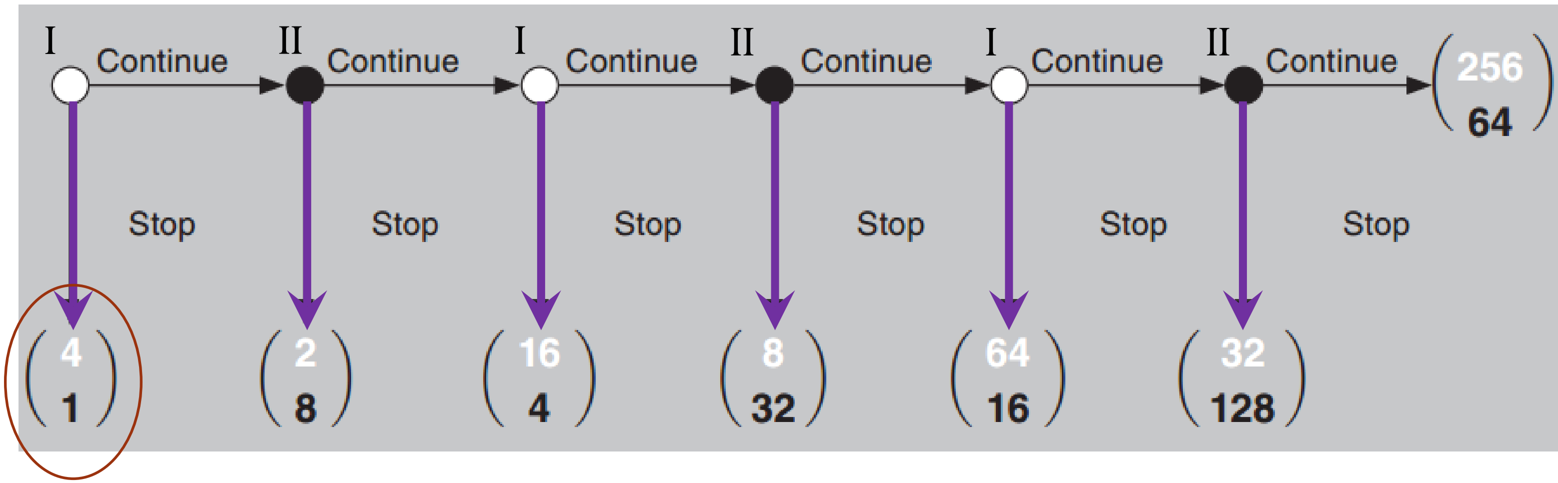
Ultimatum Game

- If each player only cares about money, what's the prediction?
  - x=99 (or even 99.99) for player A
  - But this outcome is rarely observed

- Backward induction is wrong?
  - People usually also care about fairness in such a situation
  - Once that is considered, backward induction still makes a sensible prediction
  - Suppose B will reject any offer that gives her ≥ \$30 less than A. Then what will be the prediction?

[Question: When B rejects, the game continues and it becomes B's turn to make an offer. What will be the prediction?]

Centipede Game



- This theoretical prediction is rarely observed in experiments
- Why?

TABLE 1—COLLEGE STUDENTS: PROPORTION OF OBSERVATIONS AT EACH TERMINAL NODE

	N	f_1	f_2	f_3	f_4	f_5	f_6	f_7
Panel A: UPV college students								
	40	0.075	0.150	0.350	0.300	0.100	0.025	0.000
Panel B: McKelvey and Palfrey (1992) students								
Repetitions 1–5	145	0.000	0.055	0.172	0.331	0.331	0.090	0.021
Repetitions 6–10	136	0.015	0.074	0.228	0.441	0.169	0.066	0.007
Total	281	0.007	0.064	0.199	0.384	0.253	0.078	0.014

Note: The McKelvey and Palfrey students played ten repetitions of a six-node centipede game with about one-tenth lower stakes than the game played by the Universidad del País Vasco (UPV) students, who played it just once.

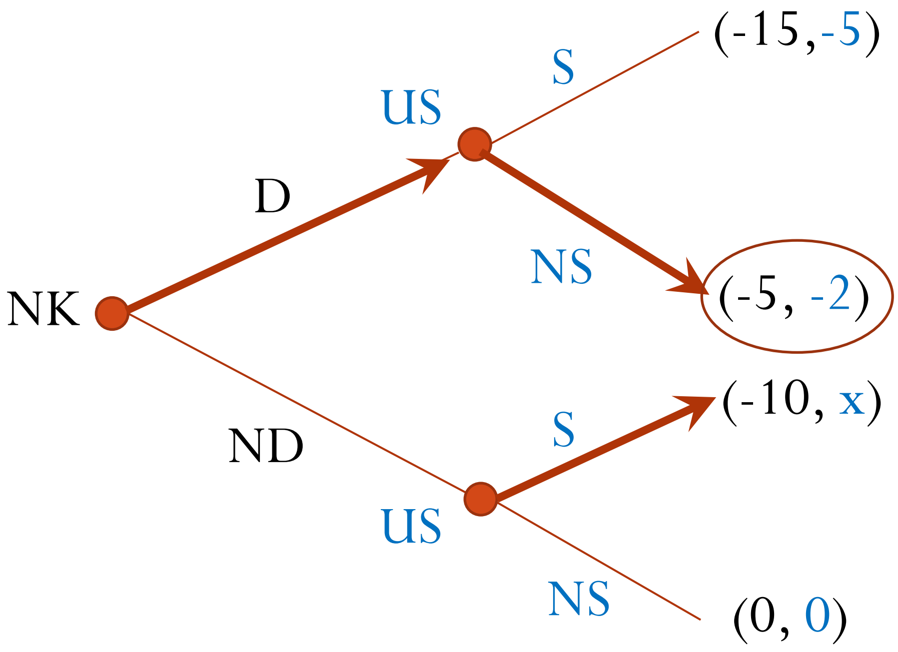
- This suggests that *limited foresight* might be the reason why ordinary players don't stop in the beginning
- This is also an example where "bounded rationality" can benefit players in a game

(Source: Palacios-Huerta and Volij (2009) in *American Economic Review*)

Example: North Korea Crisis

D: develop nuclear weapons

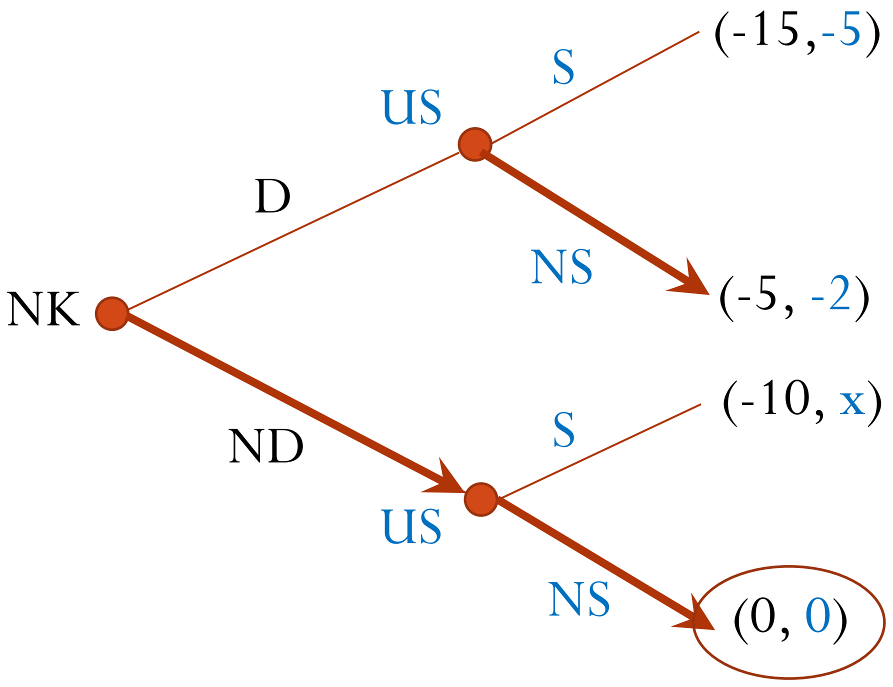
S: strike



Prediction if $x>0$?

Example: North Korea Crisis

D: develop nuclear weapons
S: strike



Prediction if $x<0$?

Reducing the US's payoff number x actually *benefits* the US !
Making $x<0$ is the same as *committing not to strike*.

Commitment

- ***Commitment***: lock yourself into a certain course of action that you might not otherwise choose
  - It can influence the game structure and thus the rivals' moves
  - In a "game", *having fewer options or making some options worse* can sometimes benefit you!

- But, commitment works only if it is *credible*

- Other examples of commitment:
  - Military stories (e.g., burning ships)
  - Wedding or engagement announcement
  - Retirement savings account

…

Summary

- Backward induction finds NE in sequential-move games

- Main games discussed:
  - *Hold-up game*: opportunism can lead to low efficiency
  - *Ultimatum game*: people often care about non-monetary things such as fairness
  - *Centipede game*: limited foresight makes a long backward induction hard
  - *North Korea crisis game*: commitment; *having fewer or worse options can sometimes help a player*